Working Paper # 30

Revision of BE Software Engineering

Sponsored by SEECS

Background

1. The School of Electrical Engineering and Computer Science (SEECS), Department of Computing (DoC) is currently offering a Bachelors of Engineering in Software Engineering (BESE) degree at the under-graduate level.

2. The BESE program at DoC spans over four years duration. Currently, it is based on the curriculum approved by the Pakistan Engineering Council (PEC) in 2020. It was approved vide WP No 3 of 61st ACM held on August 30, 2021.

Rationale

3. This document proposes to update the BESE program for the following reason:

   a. It incorporates the changes required to meet the minimum regular semester load after NUST revised the FYP's course load distribution from 3 CHs each in the 7th and 8th semesters to 2 CHs and 4 CHs, respectively.

4. Details of the modifications are presented in Para 8.

Educational Objectives

5. The aim is to align the BESE Program at SEECS NUST with PEC revised curriculum for Bachelors in Software Engineering 2020 and to improve the quality of FYP in the BESE Program at NUST as per academia and industry needs.

Additional Resources Required

6. No additional resource in terms of instructor or classroom will be required.

Proposed Time Frame

7. The proposed changes in the curriculum w.r.t to FYP credit hours modifications will be applied to all ongoing batches of BESE, i.e., intakes of Fall 2019 and onwards w.e.f Fall 2022.

Summary of the Proposed Changes

8. The summary of the proposed changes are given below.

a. **Changes in CHs Distribution for FYP (SE-498 & SE-499 – Senior Project)**. In the light of the revised FYP policy notified by NUST, the distribution of CHs for FYP has been incorporated; 2 CHs are allocated for semester-7 and 4 CHs for semester-8.

b. **Inclusion of an Elective Course for the intakes of Fall 2019 and 2020**. The following NUST approved Sociology Elective course is proposed for inclusion in the revised curriculum in 8th Semester for the intakes of Fall 2019 – 2020 to meet the minimum regular semester CHs load.

#	Course Code	Course Title	CHs	Offering in the Proposed Curriculum
1	ECO-130	Social Science Elective (Engineering Economics)	2-0	Semester 8

c. **Changes in semester-wise course offerings for intakes of Fall 2021 & onwards**

Following elective courses in 7th semester and 8th semester are swapped for the intakes of Fall 2021 & onwards to meet the minimum regular semester CHs load.

#	Course Code	Course Title	CHs	Offering in the Existing Curriculum	Offering in the Proposed Curriculum
1		Engineering Elective IV	3-1	Semester 7	Semester 8
2		MDEE-II (Internet of Things)	3-0	Semester 8	Semester 7

d. **Revised Semester-Wise Course Offerings**. The revised semester-wise course offering for Fall 2019-2020 and Fall 2021 & onwards intakes are presented in Annex-I and Annex-II, respectively.

Approval of FBS

9. Approved by the FBS held on November 16th, 2022.

Faculty Involved in the Preparation of Working Paper

10. Following faculty members participated in the process:

a. Dr Rafia Mumtaz, HoD, Tenured Associate Professor, Software Engineering Department, DoC, SEECS

b. Dr. Zuhair Zafar, Assistant Professor, Software Engineering Department, DoC, SEECS

Comments of Academics Directorate

11. No additional requirement of faculty, classrooms & labs. The proposal was deliberated and endorsed by UCRC held on 16 Dec 2022.

Recommendations of Academics Directorate

12. Revision of BE Software Engineering at SEECS and MCS is recommended for approval for intakes of Fall 2019 onwards batches w.e.f Spring 2023.

13. Academic council is requested for the decision.

Annex – I

Bachelors of Engineering in Software Engineering (For Fall 2019 – 2020 intakes) Revised Curriculum						
S. No	**Course Code**	**Subject**	**CHs Teaching**	**CHs Labs**	**Total**	**Semester**
1	HU-100	English	2	0	2	**1st**
2	CS-114	Fundamentals of Programming	2	1	3	
3	HU-107	Pakistan Studies	2	0	2	
4	MATH-101	Calculus and Analytical Geometry	3	0	3	
5	ME-105	Workshop Practice	0	1	1	
6	PHY-102	Applied Physics	2	1	3	
7	MATH-161	Discrete Mathematics	3	0	3	
Total CHs			**14**	**3**	**17**	**17**
S. No	**Course Code**	**Subject**	**CHs Teaching**	**CHs Labs**	**Total**	**Semester**
1	CS-212	Object Oriented Programming (OOP)	3	1	4	**2nd**
2	HU-101	Islamic Studies	2	0	2	
3	MATH-121	Linear Algebra and ODEs	3	0	3	
4	ME-109	Engineering Drawing	0	2	2	
5	HU-109	Communication Skills	2	0	2	
6	EE-221	Digital Logic Design	3	1	4	
Total CHs			**13**	**4**	**17**	**17**
S. No	**Course Code**	**Subject**	**CHs Teaching**	**CHs Labs**	**Total**	**Semester**
1	CS-220	Database Systems	3	1	4	**3rd**
2	SE-200	Software Engineering	3	0	3	
3	CS-250	Data Structures & Algorithms	3	1	4	
4	MATH-361	Probability and Statistics	3	0	3	
5	EE-212	Basic Electronics	2	1	3	
6	HU-210	Technical Writing	3	0	3	

Total CHs			17	3	20	20
S. No	**Course Code**	**Subject**	**CHs Teaching**	**CHs Labs**	**Total**	**Semester**
1	EE-321	Computer Architecture and Organization	3	1	4	**4th**
2	SE-311	Software Requirements Engineering	3	0	3	
3	EE-353	Computer Networks	3	1	4	
4	CS-251	Design & Analysis of Algorithm	3	0	3	
5	MATH-232	Complex Variables and Transform	3	**0**	3	
Total CHs			**15**	**2**	**17**	**17**
S. No	**Course Code**	**Subject**	**CHs Teaching**	**CHs Labs**	**Total**	**Semester**
1	CS-330	Operating Systems	3	1	4	**5th**
2	SE-210	Software Design and Architecture	3	1	4	
3	HU-223	Professional Ethics	3	0	3	
4	MGT-121	Introduction to Management (GE Elective – I)	3	0	3	
5	CS-344	Web Engineering	3	1	4	
Total CHs			**15**	**3**	**18**	**18**
S. No	**Course Code**	**Subject**	**CHs Teaching**	**CHs Labs**	**Total**	**Semester**
1	CS-370	Software Construction	3	1	4	**6th**
2	SE-320	Formal Methods	3	0	3	
3	SE-321	Software Quality Engineering	3	0	3	
4	MGT-271	Entrepreneurship	2	0	2	
5	CS-471	Machine Learning	3	1	4	
6	MATH- 351	Numerical Methods	3	0	3	
Total CHs			**17**	**2**	**19**	**19**
S. No	**Course Code**	**Subject**	**CHs Teaching**	**CHs Labs**	**Total**	**Semester**

1	SE-430	Software Project Management	3	0	3	7th
2	SE-498	Final Year Project - I	0	2	2	
3	CS-335	Internet of Things	3	0	3	
4	CS-332 CS-381 CS-473 CS-476	Elective (Distributed Computing / Network Security / Theory of Intelligent System / Speechand Image Processing)	3	1	4	
5	FIN-100	Principles of Accounting	3	**0**	3	
Total CHs			**12**	**3**	**15**	**15**
S. No	**Course Code**	**Subject**	**CHs Teaching**	**CHs Labs**	**Total**	**Semester**
1	SE-499	Final Year Project - II	0	4	4	8th
2	CS-474/ CS-481	Computer Vision / Computer Forensics	2	1	3	
3	HRM-241 HRM-442	Organizational Behavior/ Human Resource Management	3	0	3	
4	CSL-401	Community Service	1*	1*	0	
5	ECO-130	Social Science Elective (Engineering Economics)	2	0	2	
Total CHs			**7**	**5**	**12**	**12**
		Grand Total (Credit Hours)	**110**	**25**		**135**

Note:

- * Community Service is a non-credit course.

Annex –II

Bachelors of Engineering in Software Engineering (For Fall 2021 and onwards intakes) Revised Curriculum						
S. No	**Course Code**	**Subject**	**CHs Teaching**	**CHs Labs**	**Total**	**Semester**
1	HU-100	English	2	0	2	**1st**
2	CS-114	Fundamentals of Programming	2	1	3	
3	HU-107	Pakistan Studies	2	0	2	
4	MATH-101	Calculus and Analytical Geometry	3	0	3	
5	ME-105	Workshop Practice	0	1	1	
6	PHY-102	Applied Physics	2	1	3	
7	MATH-161	Discrete Mathematics	3	0	3	
8	OHS-101	Occupational Health and Safety	1	0	1	
		Total CHs	**15**	**3**	**18**	**18**
S. No	**Course Code**	**Subject**	**CHs Teaching**	**CHs Labs**	**Tota l**	**Semester**
1	CS-212	Object Oriented Programming (OOP)	3	1	4	**2nd**
2	HU-101	Islamic Studies	2	0	2	
3	MATH-121	Linear Algebra and ODEs	3	0	3	
4	ME-109	Engineering Drawing	0	2	2	
5	HU-109	Communication Skills	2	0	2	
6	EE-122	Computer Architecture and Logic Design	3	1	4	

		Total CHs	**13**	**4**	**17**	**17**
S. No	**Course Code**	**Subject**	**CHs Teaching**	**CHs Labs**	**Total**	**Semester**
1	CS-220	Database Systems	3	1	4	**3rd**
2	SE-200	Software Engineering	3	0	3	
3	CS-250	Data Structures & Algorithms	3	1	4	
4	MATH-361	Probability and Statistics	3	0	3	
5	EE-353	Computer Networks	3	1	4	
		Total CHs	**15**	**3**	**18**	**18**
S. No	**Course Code**	**Subject**	**CHs Teaching**	**CHs Labs**	**Total**	**Semester**
1	CS-330	Operating Systems	3	1	4	**4th**
2	CS-251	Design & Analysis of Algorithm	3	0	3	
3	MATH-232	Complex Variables and Transform	3	0	3	
4	SE-211	Software Design and Architecture	2	1	3	
5	MATH-352	Numerical Methods	2	1	3	
6		Sociology Elective – I (Professional Ethics)	2	0	2	
		Total CHs	**15**	**3**	**18**	**18**
S. No	**Course Code**	**Subject**	**CHs Teaching**	**CHs Labs**	**Total**	**Semester**
1		Management Science Elective – I (Engg. Management / Project Management)	3	0	3	**5th**
2		Engineering Elective – I (Web Engineering)	3	1	4	
3	SE-315	Cloud Computing	2	1	3	
4		MDEE – I (Basic Electronics / Embedded Systems)	2	1	3	

5	HU-212	Technical & Business Writing	2	0	2	
6	SE-314	Software Construction	2	1	3	
		Total CHs	**14**	**4**	**18**	**18**
S. No	**Course Code**	**Subject**	**CHs Teaching**	**CHs Labs**	**Total**	**Semester**
1	SE-320	Formal Methods	3	0	3	**6th**
2	SE-321	Software Quality Engineering	3	0	3	
3		Management Science Elective – II (Entrepreneurship)	2	0	2	
4		Engineering Elective-II (Distributed Computing /Information Retrieval)	3	X	3+X	
5	SE-430	Software Project Management	3	0	3	
6	CS-261	Human Computer Interaction	2	1	3	
Total CHs			**16**	**1+X**	**17+X**	**17+X**
S. No	**Course Code**	**Subject**	**CHs Teaching**	**CHs Labs**	**Total**	**Semester**
1	SE-498	Final Year Project – I	0	2	2	**7th**
2		Engineering Elective – III (DIP / Network Security)	3	X	3+X	
3		MDEE – II (Internet of Things)	3	0	3	
4		Sociology Elective – II (Engg. Economics)	2	0	2	
5	CS-484	Information Security	3	0	3	
		Total CHs	**11**	**2+X**	**13+X**	**13+X**
S. No	**Course Code**	**Subject**	**CHs Teaching**	**CHs Labs**	**Total**	**Semester**
1	SE-499	Final Year Project – II	0	4	4	**8th**
2		Engineering Elective – IV (Data Warehousing and Data Mining)	3	1	4	
3	CSL-401	Community Service	1*	1*		

4		Engineering Elective – V (Computer Forensics / Artificial Intelligence)	3	1	4	
		Total CHs	**6**	**6**	**12**	**12**
		Grand Total (Credit Hours)	**105**	**26+X**		**131+X**

Note:

* Community Service is a non-credit course.